VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi,-590014, Karnataka

Project Phase I Report On

"Intelligent Cyber Threat Prediction Model for Blockchain-Enabled Land Revenue Records"

Submitted in the partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF ENGINEERING

IN

INFORMATION SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the project phase I, Project titled as **"Intelligent Cyber Threat Prediction Model for Blockchain-Enabled Land Revenue Records**" presented by **Ananya Kamath S(1MJ23IS006), Imitiyaz Ali(1MJ23IS029),Mamatha H V(1MJ24IS420),Manikanta S(1MJ24IS421)**Bonafide students of **MVJ College of Engineering**, Bengaluru, in partial fulfilment for the award of **Bachelor of Engineering** in **Information Science & Engineering** of **Visvesvaraya Technological University**, Belagavi during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated have been incorporated in the report and submitted in the department library. The project phase 1 has been approved as it satisfies the academic requirements in respect of the Project phase 1 prescribed by the institution for the said degree.

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DECLARATION

We, Ananya Kamath s (1MJ23IS006), Imitiyaz Ali (1MJ23IS029), Mamatha H V(1MJ24IS420), Manikanta S (1MJ24IS421) students of seventh semester B.E., Department of Information Science and Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the Project phase -1 titled " Intelligent Cyber Threat Prediction Model for Blockchain-Enabled Land Revenue Records " has been carried out by us and submitted in partial fulfilment for the award of Degree of Bachelor of Engineering in Information Science and Engineering during the academic year 2025-26

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

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ABSTRACT

The rapid digitization of land revenue record systems has significantly improved efficiency, transparency, and accessibility in managing land-related data. However, this transformation has also introduced serious cybersecurity challenges such as unauthorized access, data breaches, data tampering, and ransomware attacks. These threats pose significant risks to the integrity and reliability of land records, which are critical for property ownership and legal verification. Traditional security mechanisms are often insufficient to handle such evolving cyber threats, creating the need for more advanced and intelligent security solutions.

To address these challenges, this project proposes an Intelligent Cyber Threat Prediction Model integrated with blockchain technology. The system combines the security benefits of blockchain, such as decentralization and immutability, with machine learning techniques for detecting anomalies and predicting cyber threats. It includes modules for data processing, threat detection, secure storage, and real-time alert generation. This integrated approach enhances data security, ensures transparency, and enables proactive threat detection, making it a reliable and scalable solution for modern digital land record management systems.

ACRONYMS

Acronym	Abbreviation
USN	University Seat Number
SEM	Semester
AI	Artificial Intelligence
ML	Machine Learning
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
UI	User Interface
DFD	Data Flow Diagram
IDE	Integrated Development Environment
KNN	K-Nearest Neighbors
B/S	Browser/Server
DB	Database
IP	Internet Protocol

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CHAPTER 1

INTRODUCTION

1.1Aim

The aim of this project is to design and develop a secure and intelligent system for managing land revenue records using advanced technologies like Blockchain and Machine Learning. The system focuses on protecting sensitive land data from unauthorized access, cyber attacks, and data tampering. Blockchain technology is used to ensure that the stored data is immutable, transparent, and decentralized, which increases trust and reliability. Machine Learning algorithms are used to monitor system activities, detect unusual patterns, and predict possible cyber threats in advance. The project also aims to provide real-time alerts to administrators for quick response to suspicious activities. Overall, the aim is to build a system that enhances data security, ensures integrity, and improves the efficiency of land record management.

1.2 Motivation

The motivation behind this project comes from the increasing challenges faced in traditional and digital land record systems. There are many cases of data breaches, hacking, and unauthorized modifications of sensitive land information. These issues lead to fraud, disputes, and loss of trust among users. Existing systems lack proper security mechanisms and do not provide real-time monitoring or early detection of threats. With the rapid growth of technology, there is a need for a smarter and more secure system. Blockchain offers a reliable way to store data securely, while Machine Learning provides the ability to detect and predict cyber threats. This project is motivated by the need to create a system that ensures transparency, security, and trust in land record management

1.3 Problem Statement

The current land record management systems suffer from several security and reliability issues. Most systems are centralized, making them vulnerable to cyber attacks and data breaches. There is no proper mechanism to prevent data tampering or unauthorized access. Additionally, existing systems lack intelligent monitoring capabilities to detect suspicious activities in real time. This results in delayed responses to cyber threats and increases the risk of data loss and fraud. The absence of transparency and trust further complicates the system.

Therefore, there is a need for a secure, decentralized, and intelligent system that can protect data, detect threats early, and ensure reliable land record management.

1.3.1 Problems Faced

The existing systems face multiple challenges that affect their performance and reliability. One of the major issues is unauthorized access, where attackers can gain access to sensitive data. Data tampering is another serious problem, where records can be altered or deleted without proper authorization. There is also a lack of transparency, making it difficult for users to verify the authenticity of data. Another issue is the absence of real-time monitoring and alert systems, which delays the detection of cyber threats. Additionally, centralized systems suffer from scalability issues and are more prone to failure. These problems highlight the need for a more secure and intelligent solution.

1.4 Existing System

The existing land record management systems are primarily based on centralized databases where all data is stored in a single location. These systems provide basic functionalities such as data storage, retrieval, and user access. However, they have several limitations. Since the system is centralized, it creates a single point of failure, making it highly vulnerable to cyber attacks. Data can be easily modified or deleted by unauthorized users or insiders. There is no mechanism to track changes effectively or ensure data integrity. Moreover, these systems lack advanced features such as real-time threat detection and predictive analysis. As a result, the existing systems are less secure, less reliable, and prone to data breaches.

1.5 Proposed System

The proposed system introduces a secure and intelligent approach to land record management by integrating Blockchain and Machine Learning technologies. Blockchain is used to store data in a decentralized and tamper-proof manner, ensuring that once data is recorded, it cannot be altered. This improves transparency, security, and trust. Machine Learning algorithms are used to analyze system behavior and detect anomalies or suspicious activities. The system can predict potential cyber threats and generate real-time alerts for administrators. This enables quick response and prevents major security issues. The proposed system overcomes the limitations of existing systems by providing security.

CHAPTER 2

LITERATURE SURVEY

2.1 A Secure Land Record Management System Using Blockchain Technology

Authors: D. Kumar et al.

Year: 2023

This paper proposes a secure system for managing land records using blockchain technology. The system ensures data integrity, transparency, and resistance to tampering by storing records on a distributed ledger. It improves trust and reduces fraud in land transactions. The system also provides secure access to authorized users. However, it does not include real-time monitoring or predictive capabilities for detecting cyber threats. The absence of intelligent security mechanisms limits its effectiveness in preventing cyber attacks. Thus, while the system provides strong data protection, it does not offer proactive threat detection.

2.2 Blockchain and Smart Contract for Land Registration Using Ethereum Network

Authors: B. Beachwear et al.

Year: 2022

This paper presents a blockchain-based approach for secure land registration using the Ethereum network. The system utilizes smart contracts to automate the validation and execution of land transactions, thereby eliminating the need for intermediaries and reducing human errors. Smart contracts ensure that transactions are executed only when predefined conditions are satisfied, which enhances trust and efficiency in the system. The decentralized nature of blockchain ensures that land records are stored across multiple nodes, making them highly secure and resistant to tampering. Each transaction is recorded in a block and linked to previous records, ensuring immutability and transparency. The system also improves accessibility by allowing authorized users to view and verify land ownership details easily. However, the proposed model mainly focuses on data security and transaction management after data is recorded. It does not provide mechanisms for monitoring system activities or predicting cyber threats. Therefore, while the approach enhances transparency and reduces fraud, it does not fully address cybersecurity challenges.

2.3 Blockchain Infused Decentralized Land Registry System

Authors: A. Singh et al.

Year: 2022

This research introduces a decentralized land registry framework that uses blockchain technology to eliminate centralized control and improve data security. The system ensures that all records are stored in a distributed manner, making them resistant to tampering and unauthorized access. It enhances transparency by allowing users to verify transactions and ownership details. The decentralized nature of the system also improves reliability and reduces the risk of system failure. However, the system primarily focuses on data storage and does not incorporate mechanisms for detecting cyber threats. It lacks intelligent monitoring systems and real-time alert generation, which are essential for preventing cyber-attacks. Thus, the system provides strong data security but limited threat detection capabilities.

2.4 Blockchain Land Transfers: Technology, Promises, and Perils

Authors: J. Smith et al.

Year: 2022

This paper provides an overview of blockchain technology in land transfer systems, discussing its advantages and challenges. It highlights the benefits of decentralization, transparency, and security in managing land records. The study also addresses potential risks such as cyber vulnerabilities, implementation challenges, and scalability issues. It emphasizes that while blockchain enhances data integrity, it does not eliminate the need for additional security mechanisms. The paper suggests the integration of intelligent systems for monitoring and detecting cyber threats. This highlights the importance of combining blockchain with machine learning to achieve a comprehensive security solution.

2.5 Land Registry Using Blockchain

Authors: S. Sharma et al.

Year: 2021

This study proposes a decentralized land registry system that leverages blockchain technology to improve transparency and security in land record management. The system replaces traditional centralized databases with a distributed ledger, ensuring that data cannot be altered without consensus. Each transaction is validated by network participants, which enhances trust and prevents unauthorized modifications.

The system reduces fraud, corruption, and duplication of records by maintaining a single source of truth. It also enables easy verification of land ownership and transaction history. Despite these advantages, the system lacks mechanisms for real-time monitoring and detection of suspicious activities. It does not incorporate machine learning or intelligent systems to identify anomalies or predict cyber threats. As a result, the system is secure but not proactive in handling cybersecurity risks.

2.6 Online Land Registration Using Blockchain

Authors: R. Kumar et al.

Year: 2021

This paper focuses on developing an online platform for land registration using blockchain technology. The system enables users to perform land transactions digitally, eliminating the need for manual paperwork and reducing delays. Blockchain ensures that all records are securely stored and cannot be modified once recorded. The system improves efficiency, transparency, and reliability in land registration processes. It also reduces the chances of fraud and disputes by maintaining accurate records. However, the system does not include advanced security features such as intrusion detection or threat prediction. It lacks real-time monitoring capabilities and cannot identify malicious activities before they occur. Therefore, while the system enhances operational efficiency, it does not fully address cybersecurity challenges.

2.7 Blockchain-Based Framework for Secure and Reliable Land Registry System

Authors: K. Reddy et al.

Year: 2021

This study proposes a blockchain-based framework designed to enhance the security and reliability of land registry systems. The system uses encryption and decentralized storage to protect sensitive data from unauthorized access. It ensures data integrity by preventing any modification of records without proper authorization. The framework also improves transparency and accountability in land transactions. However, it does not provide solutions for detecting or predicting cyber threats. The absence of machine learning or intelligent monitoring systems limits its ability to handle advanced cyber-attacks. Thus, while the system improves data security, it lacks proactive threat detection capabilities.

2.8 Query Optimization in a Blockchain-Based Land Registry Management System

Authors: L. Wang et al.

Year: 2021

This paper addresses performance issues in blockchain-based land registry systems by optimizing query processing. The proposed approach improves the speed and efficiency of retrieving data from the blockchain. It enhances system performance and reduces response time, making it more practical for real-world applications. However, the study focuses mainly on performance improvement and does not address security concerns. It does not include mechanisms for detecting or preventing cyber threats. Therefore, while the system is efficient, it lacks comprehensive security features

2.9 Blockchain-Based Reliable Framework for Land Registration Information System

Authors: M. Patel et al.

Year: 2020

This paper presents a reliable framework for managing land registration information using blockchain technology. The system ensures secure storage, transparency, and efficient management of land records. By using cryptographic techniques, it protects data from unauthorized access and ensures integrity. The system also improves trust among users by providing a tamper-proof record of transactions. However, the framework mainly focuses on maintaining data security and does not include predictive analytics or anomaly detection mechanisms. It lacks the ability to monitor system behavior and identify potential threats in real time. Therefore, the system is secure but not intelligent in handling cyber risks.

2.10 Land Registry System Using Smart Contracts

Authors: P. Verma et al.

Year: 2020

This research focuses on using smart contracts to automate land registry processes. The system ensures that transactions are executed securely and transparently without the need for intermediaries. Smart contracts reduce delays and improve efficiency by automating verification processes. The system also minimizes human errors and enhances trust among users. However, it lacks real-time monitoring and does not include mechanisms for detecting cyber threats. It does not analyze system behavior or identify anomalies, making it vulnerable to cyber attacks. Therefore, while the system improves efficiency, it does not provide complete security.

Summary of the literature survey

From the above literature survey, it is observed that most existing systems focus on secure storage and transparency using blockchain technology. However, they lack intelligent mechanisms for predicting and preventing cyber threats. This highlights the need for an integrated system that combines blockchain with machine learning to provide real-time monitoring, anomaly detection, and proactive cybersecurity solutions.

CHAPTER 3

SYSTEM REQUIREMENTS AND SPECIFICATION

3.1 HARDWARE REQUIREMENTS

- Computer/Laptop: A system with a minimum of 8 GB RAM is required to handle data processing, machine learning model execution, and blockchain operations efficiently.
- Processor: An Intel i5 processor or higher is recommended to ensure faster computation and smooth performance during model training and system execution.
- Storage: A minimum of 256 GB hard disk space is required to store datasets, system files, and blockchain records.
- Internet Connection: A stable internet connection is necessary for accessing datasets, interacting with blockchain networks, and supporting real-time system operations.

3.2 SOFTWARE REQUIREMENTS

- Programming Language: Python is used for implementing machine learning algorithms and backend functionalities due to its simplicity and extensive library support.
- Machine Learning Libraries: Libraries such as Scikit-learn and TensorFlow are used for building, training, and evaluating machine learning models.
- Blockchain Platform: Ethereum or Hyperledger is used to implement blockchain technology for secure storage and management of land records.
- Database: MySQL or MongoDB is used to store system data and manage records efficiently.
- Development Tools: IDEs such as Visual Studio Code, PyCharm, or Jupyter Notebook are used for coding, testing, and debugging the system.
- Frontend Technologies: HTML, CSS, and JavaScript are used to design the user interface and ensure a user-friendly experience.
- Operating System: The system can be developed and executed on Windows or Linux operating systems.

3.3 FUNCTIONAL REQUIREMENTS

- User Registration and Login: The system should allow users to register and log in securely using authentication mechanisms.
- Land Record Management: The system must enable users to upload, view, and manage land records efficiently.
- Secure Data Storage: All land records should be stored securely using blockchain technology to ensure immutability and integrity.
- Cyber Threat Detection: The system should analyze user activities and detect potential cyber threats using machine learning algorithms.
- Alert and Notification System: The system must generate real-time alerts and notifications when suspicious activities are detected.
- Admin Dashboard: An administrator interface should be provided to monitor system activities, manage records, and respond to alerts.
- Data Processing: The system should process and analyze large volumes of data efficiently for accurate prediction.

3.4 NON-FUNCTIONAL REQUIREMENTS

- Security: The system must provide high-level security to protect sensitive land records from unauthorized access and cyber threats.
- Performance: The system should respond quickly and handle multiple operations efficiently without delays.
- Scalability: The system must be capable of handling large volumes of data and increasing user demands.
- Reliability: The system should function consistently without failures and ensure continuous availability.
- Data Integrity: The system must ensure that data remains accurate and cannot be altered without authorization.
- Usability: The system should have a simple and user-friendly interface for easy interaction.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system architecture represents the overall structure of the proposed Intelligent Cyber Threat Prediction Model for blockchain-enabled land revenue records. It illustrates how different components interact to ensure secure data processing, threat detection, and storage.

The process begins with the user, who interacts with the system through a web interface developed using frontend technologies such as HTML, CSS, and JavaScript. The user can perform operations such as uploading land records, accessing data, and initiating transactions.

The input data is then passed to the data processing module, where preprocessing operations such as cleaning, normalization, and transformation are performed. This ensures that the data is suitable for analysis by machine learning algorithms.

The processed data is then analyzed by the machine learning module, which uses algorithms such as Random Forest and Decision Tree to detect anomalies and predict potential cyber threats. Based on the analysis, the system determines whether the activity is normal or suspicious.

If any suspicious activity is detected, the alert system generates real-time notifications to inform administrators, enabling them to take immediate action. If no threat is detected, the data is securely stored in the blockchain.

The blockchain module ensures that all land records are stored in a decentralized and tamper-proof manner. Each transaction is recorded as a block and linked to previous records, ensuring data integrity and transparency.

Finally, the database stores system-related information such as user data, logs, and transaction records. The overall architecture ensures secure data flow, efficient processing, and proactive threat detection.

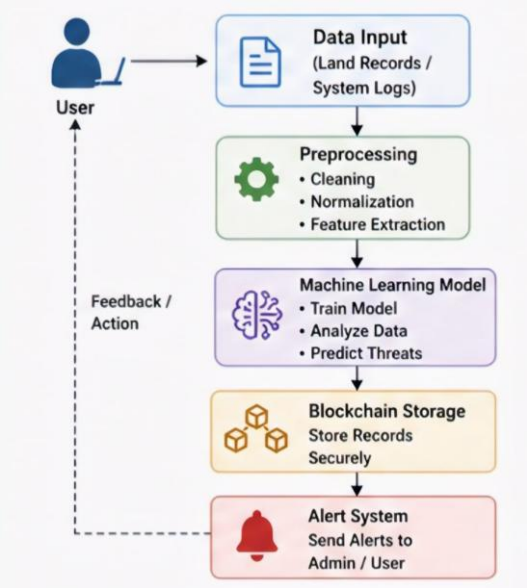


Fig 4.1 : System architecture

4.2 FLOWCHART

The flowchart illustrates the step-by-step working process of the proposed system. It provides a clear understanding of how data flows through different stages, from user input to threat detection and secure storage.

The process begins with user login, where authorized users access the system. After successful authentication, the user inputs data such as land records or transaction details.

The input data is then processed in the data preprocessing stage, where noise, missing values, and irrelevant information are removed. This is followed by feature extraction, where important features are selected for analysis.

The processed data is then passed to the machine learning model, which analyzes the data and identifies patterns. The model determines whether the activity is normal or indicates a potential cyber threat. If a threat is detected, the system generates an alert and notifies the administrator for immediate action. If no threat is detected, the data is securely stored in the blockchain to ensure integrity and transparency.

The process ends after the data is either stored or an alert is generated. This flowchart helps in understanding the logical sequence of operations in the system.

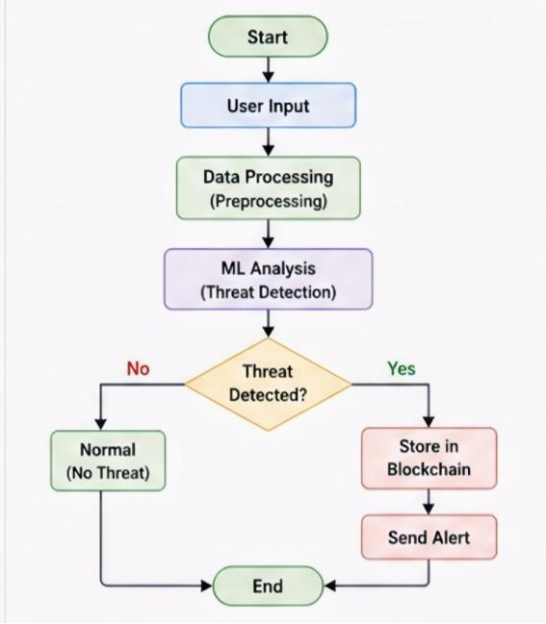


Fig 4.2 : flowchart

4.3 DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram (DFD) represents the flow of data within the system and explains how input data is transformed into meaningful output through various processes. It helps in understanding how information moves between the user, system modules, and storage components.

4.3.1 DFD LEVEL 0

The Level 0 DFD provides a high-level view of the system as a single process. It shows the interaction between the external entity (User) and the system.

The user provides input data such as land records and transaction details to the system. The system processes this data using machine learning algorithms and blockchain technology. The output is either a threat alert or securely stored records.

Main Components:

- User (External Entity)
- Cyber Threat Prediction System (Process)
- Output (Alert / Stored Data)

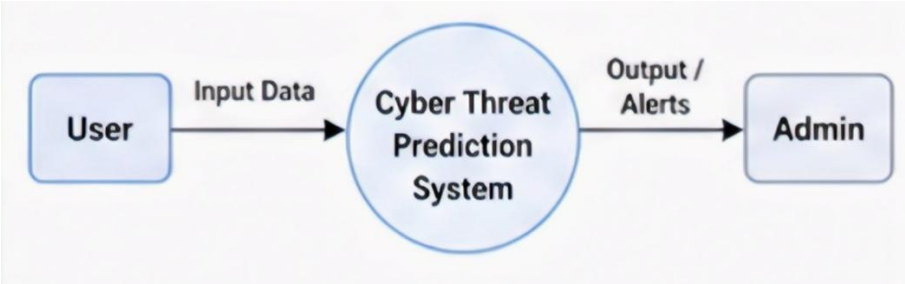


Fig 4.3: DFD Level 0 Diagram

4.3.2 DFD LEVEL 1

The Level 1 DFD provides a more detailed view of internal system processes. It breaks the system into multiple sub-processes.

Processes included:

1. Data Input
2. Data Preprocessing
3. Feature Extraction
4. Machine Learning Analysis
5. Decision Making
6. Blockchain Storage
7. Alert Generation

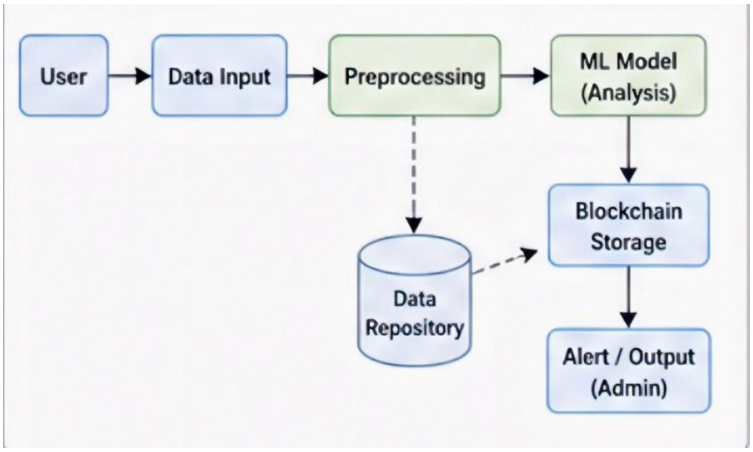


Fig 4.4: DFD Level 1 Diagram

4.4 USE CASE DIAGRAM

The Use Case Diagram represents the interaction between users and the system functionalities. It defines how different actors use the system and what operations they can perform.

Actors:

- User
- Admin

Use Cases:

- Login
- Upload Land Records
- View Records
- Detect Cyber Threat
- Receive Alerts
- Store Data in Blockchain
- Monitor System Activities

Description:

The user interacts with the system to upload and view land records, while the admin monitors system activities and receives alerts regarding suspicious behavior.

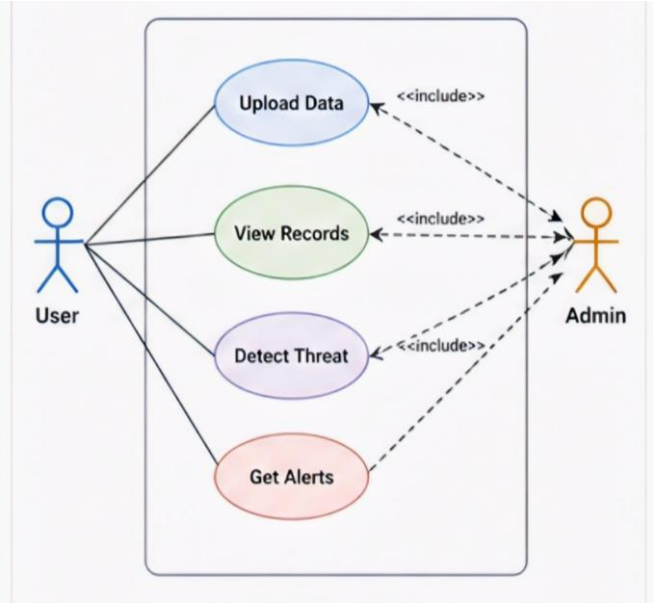


Fig 4.5: Use Case Diagram

4.5 ACTIVITY DIAGRAM

The Activity Diagram represents the dynamic workflow of the system. It shows the sequence of activities performed from the start to the end of the process.

Workflow:

Start → Login → Input Data → Data Preprocessing → Feature Extraction → ML Analysis → Decision Making → Alert Generation / Blockchain Storage → End

Explanation:

If the system detects suspicious activity, an alert is generated. Otherwise, the data is securely stored in the blockchain network.

This diagram helps in understanding the step-by-step execution flow of the system.

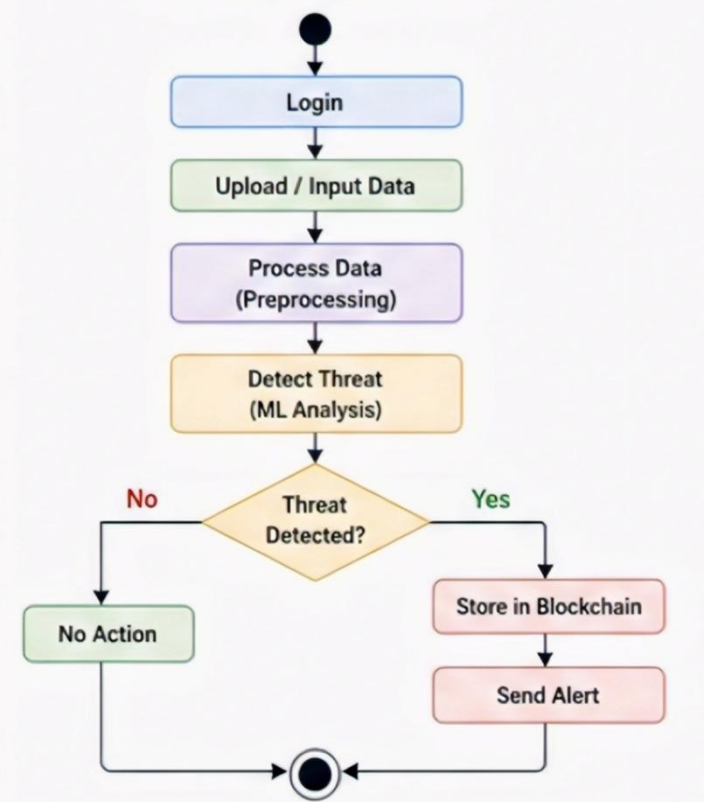


Fig 4.6: Activity Diagram

4.6 SEQUENCE DIAGRAM

The Sequence Diagram shows the interaction between different components of the system in chronological order. It represents how messages are exchanged over time.

Components:

- User
- System Interface
- Machine Learning Model
- Blockchain Module
- Admin

Sequence Flow:

1. User sends data to system
2. System forwards data to ML model
3. ML model analyses data and returns result
4. System sends safe data to blockchain
5. If threat detected, system sends alert to admin

This diagram clearly shows the time-based interaction between system components.

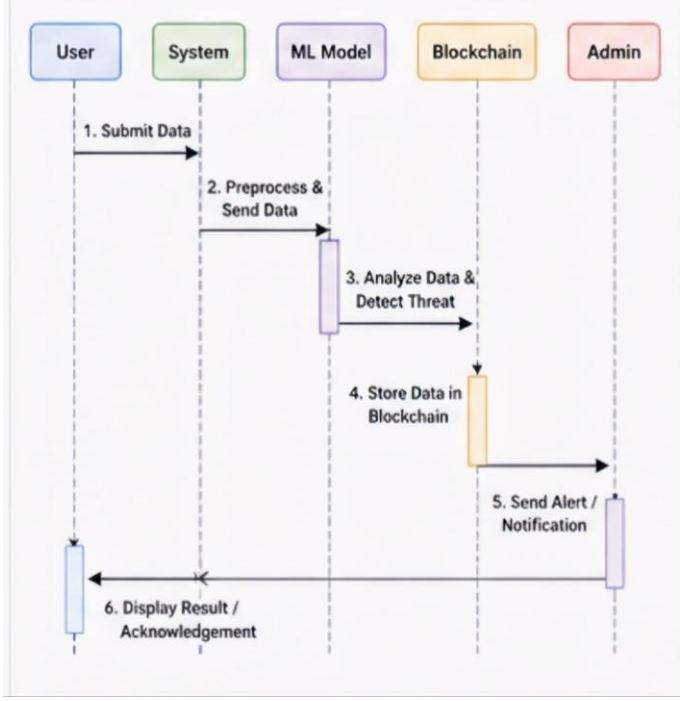


Fig 4.7: Sequence Diagram

4.7 BLOCKCHAIN STRUCTURE DIAGRAM

The Blockchain Structure Diagram represents how data is stored in a secure and tamper-proof manner using blocks.

Structure of a Block:

Each block contains:

- Data (Land record / transaction)
- Hash (Unique identifier)
- Previous Hash (Link to previous block)
- Timestamp

Working:

Blocks are linked sequentially forming a chain. Any change in one block affects all subsequent blocks, ensuring data integrity and security.

Advantages:

- Immutable records
- High security
- Transparent transactions
- Fraud prevention

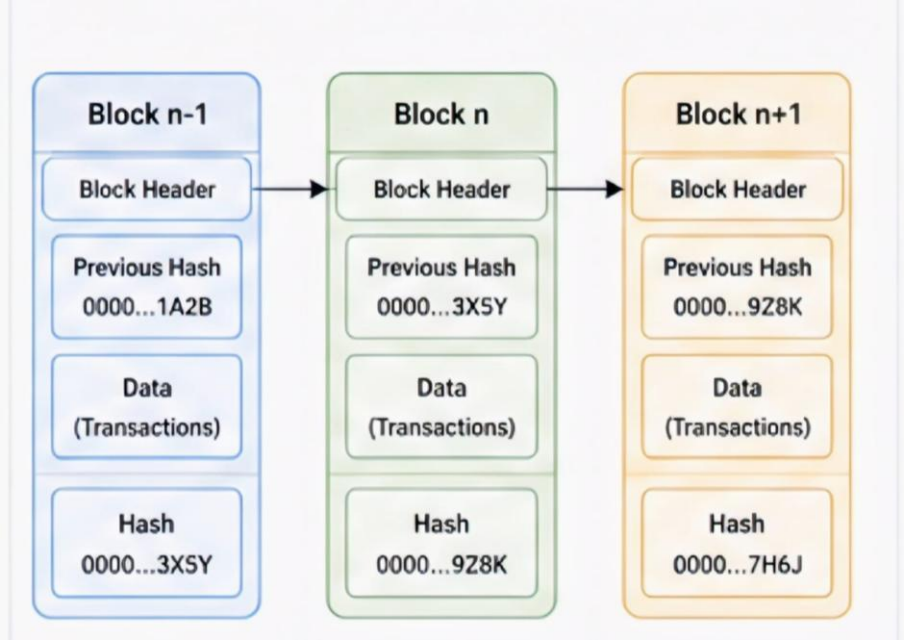


Fig 4.8: Blockchain Structure Diagram

4.8 DECISION TREE / ALGORITHM WORKING DIAGRAM

The Decision Tree Diagram represents the working logic of the machine learning model used for cyber threat detection.

Decision Flow:

- Is login valid?
- Yes → Check transaction behaviour
- Normal → Store in blockchain
- Abnormal → Generate alert
- No → Immediate alert

Explanation:

The decision tree helps the system classify activities as normal or suspicious based on learned patterns from the dataset. It improves detection accuracy and reduces false predictions.

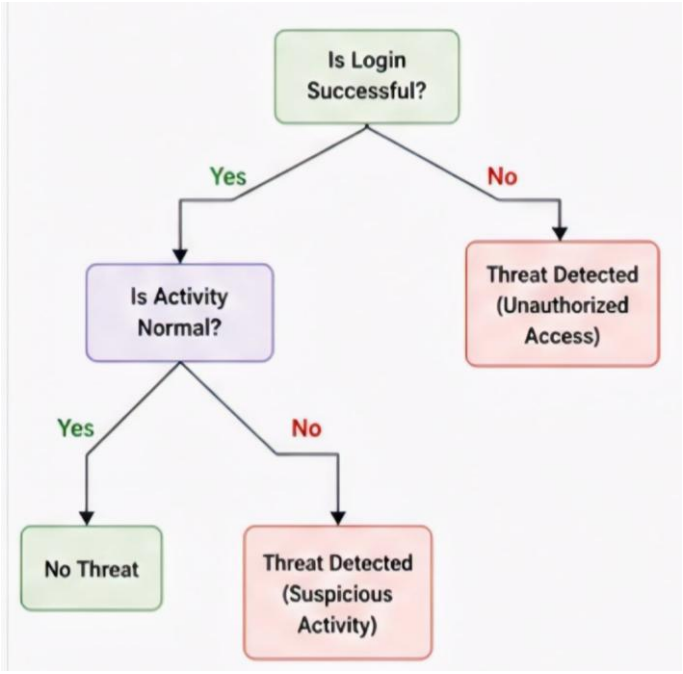


Fig 4.9: Decision Tree Diagram

The methodology of the proposed system clearly defines the structured approach used to design and implement the Intelligent Cyber Threat Prediction Model integrated with Blockchain technology. The system begins with data collection and preprocessing, followed by feature extraction and machine learning model development.

The trained model effectively detects cyber threats in real time by analyzing user behavior and system activities. Blockchain technology ensures secure and tamper-proof storage of land revenue records, maintaining transparency and integrity.

The integration of machine learning and blockchain provides a highly secure, efficient, and intelligent system for protecting sensitive land records from cyber threats. Continuous monitoring and alert generation further enhance system reliability and responsiveness.

CHAPTER 5

METHODOLOGY

The methodology of the proposed system describes the step-by-step process involved in designing and developing an Intelligent Cyber Threat Prediction Model integrated with blockchain technology. The system aims to ensure secure storage of land revenue records while enabling proactive detection and prediction of cyber threats. The methodology includes various stages such as data collection, preprocessing, feature extraction, model development, blockchain integration, and alert generation.

5.1 DATA COLLECTION

The first step in the methodology is data collection. In this stage, relevant datasets are gathered to train and test the machine learning models. The system collects cybersecurity datasets that include information about different types of cyber attacks such as ransomware, malware, phishing, and intrusion activities. These datasets contain features such as login attempts, IP addresses, transaction patterns, and system logs.

In addition to cybersecurity datasets, sample land revenue record data is also collected. This includes details related to land ownership, transactions, user activities, and system access patterns. Collecting both types of data helps in building a comprehensive model that can analyze system behavior and detect potential threats effectively.

5.2 DATA PREPROCESSING

The collected data is often raw and may contain missing values, duplicates, and irrelevant information. Therefore, data preprocessing is performed to clean and prepare the data for further analysis. In this stage, missing values are handled using appropriate techniques such as replacement or removal. Duplicate entries are eliminated to avoid redundancy.

Data normalization and transformation are also performed to ensure that all features are in a consistent format. This improves the performance of machine learning algorithms. Noise and irrelevant features are removed to enhance the accuracy of the model. Preprocessing ensures that the dataset is clean, structured, and suitable for

analysis.

5.3 FEATURE SELECTION AND EXTRACTION

Feature selection is an important step in building an effective machine learning model. In this stage, relevant features that contribute to identifying cyber threats are selected from the dataset. These features may include login frequency, unusual transaction patterns, access time, IP address behavior, and system usage patterns.

Feature extraction techniques are used to derive new features that improve the model's ability to detect anomalies. By selecting the most relevant features, the system reduces complexity and improves accuracy. This step plays a crucial role in enhancing the performance of the prediction model.

5.4 MACHINE LEARNING MODEL DEVELOPMENT

In this stage, machine learning algorithms are applied to develop the cyber threat prediction model. Algorithms such as Decision Tree, Random Forest, and Logistic Regression are used to analyze the dataset and identify patterns associated with cyber attacks.

The dataset is divided into training and testing sets. The training set is used to train the model, while the testing set is used to evaluate its performance. The model learns from historical data and identifies patterns that indicate normal and abnormal behavior. This enables the system to detect anomalies and predict potential threats.

5.5 MODEL TRAINING AND EVALUATION

After selecting the appropriate algorithms, the model is trained using the prepared dataset. During training, the model learns relationships between different features and their impact on cyber threats. Once training is complete, the model is tested using unseen data to evaluate its performance.

Evaluation metrics such as accuracy, precision, recall, and confusion matrix are used to measure the effectiveness of the model. These metrics help in determining how well the model can detect cyber threats and minimize false predictions. Based on the evaluation results, the model can be fine-tuned to improve performance.

5.6 THREAT PREDICTION AND DETECTION

The trained model is then used to predict cyber threats in real time. The system continuously monitors user activities, transactions, and system behavior. If any unusual pattern or anomaly is detected, it is identified as a potential threat.

The model classifies activities as normal or suspicious based on learned patterns. This enables early detection of cyber attacks such as unauthorized access, data tampering, and ransomware. The ability to predict threats before they occur helps in preventing potential damage.

5.7 BLOCKCHAIN INTEGRATION

Blockchain technology is integrated into the system to ensure secure storage and management of land revenue records. The blockchain network stores all transaction data in a decentralized and tamper-proof manner. Each record is stored in a block and linked to previous blocks, ensuring immutability.

Smart contracts are used to validate and manage transactions automatically. This eliminates the need for intermediaries and reduces the risk of fraud. Blockchain ensures that data cannot be altered without proper authorization, thereby maintaining integrity and transparency.

5.8 ALERT GENERATION AND SYSTEM MONITORING

The final stage of the methodology involves generating alerts and monitoring system activities. When the machine learning model detects suspicious behavior, the system generates real-time alerts to notify administrators. These alerts help in taking immediate action to prevent cyber attacks.

The system also provides a dashboard for monitoring activities and analyzing system performance. Continuous monitoring ensures that the system remains secure and efficient. This stage plays a crucial role in maintaining the reliability and effectiveness of the proposed system.

Overall, the methodology integrates machine learning and blockchain technologies to create a secure, intelligent, and efficient land record management system. The combination of proactive threat detection and

secure data storage ensures improved security and reliability.

ALGORITHM

Input:

User data, land record details, and system activity logs

Output:

Securely stored records in blockchain + alert messages if threat is detected

Step 1: Start the System

Initialize the web application

Connect to database and blockchain network

Load machine learning model

Step 2: User Input

User enters land record data or system activity

Data is sent to the backend system

Step 3: Data Preprocessing

Clean the data

Remove noise and missing values

Convert data into suitable format for ML model

Step 4: Feature Extraction

Identify important features related to cyber threats

Prepare dataset for analysis

Step 5: Machine Learning Analysis

Apply algorithms (Random Forest, KNN)

Analyze data for unusual patterns

Step 6: Threat Detection

Check whether the activity is normal or anomalous

Step 7: If Threat Detected

Store the record securely in blockchain

Generate alert notification

Send alert to admin/dashboard

Step 8: If No Threat

Mark activity as normal

Continue system operation

Step 9: Monitoring and Logging

Store logs for future analysis

Enable real-time monitoring

Step 10: End

Display result (safe / threat detected)

System continues running

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

The proposed Intelligent Cyber Threat Prediction Model integrated with blockchain technology provides a secure and efficient solution for managing digital land revenue records. By combining machine learning with blockchain, the system ensures data integrity, transparency, and protection against tampering while enabling proactive detection of cyber threats. The model analyzes system behavior, detects anomalies, and generates real-time alerts, thereby reducing the risk of unauthorized access, data breaches, and fraud. Overall, the system enhances reliability and builds trust in digital land record management.

Future enhancement

In the future, the system can be further enhanced by incorporating advanced deep learning techniques for improved threat prediction accuracy and adaptive learning. Integration with cloud computing and IoT-based monitoring can enable better scalability and real-time data analysis. Additionally, implementing multi-factor authentication and biometric security can strengthen access control. The system can also be extended to support nationwide land record integration, making it more robust, scalable, and suitable for large-scale real-world applications..

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